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# What determines whether a tumour cell depends on KMT2D

An independent computational project · public data only · every analysis pre-registered before it was run

> **Scale.** 19 KMT2D perturbation contrasts from 14 GEO series — 99 samples, 9 independent study groups, 6 tissues — integrated with DepMap 26Q3 CRISPR screens across 1,361 cell lines. Python package with 143 tests and continuous integration.

## FINDINGS

- **The fitness requirement is broad; colorectal is an ordinary lineage.** Across 20 lineages with 10,000 permutations each, Bowel sits at −0.03 against a rest-mean of −0.41 (n=85, q=0.65) — ninth of twenty. Most dependent: kidney and myeloid −0.19, ovarian −0.18, breast −0.14. Least: soft tissue +0.25, bone +0.14, brain +0.10.
- **Dependency is set by differentiation state, not by lineage or by dosage.** KMT2D's own damaging-mutation status does not predict its dependency (p=0.26). Genome-wide, 734 of 19,212 genes clear q<0.05 but the strongest |r| anywhere is **0.199** — a real, reproducible axis far too weak to stratify on, reported as the finding rather than dressed up as a signature. With lineage residualised out, both tails are one thing: epithelial splicing and apical junction genes against a mesenchymal set.
- **The two data types meet.** A gene set defined *only* by the CRISPR screens falls relative to its counterpart when KMT2D is removed: −0.17 over 9 study-group means, 7 of 9 negative, permutation p=0.0001, expression-matched competitive p=0.001. Direction pre-registered.
- **Positive control resolves complex identity.** Of 17,689 genes, KDM6A ranks 1 (r=+0.51) and NCOA6 ranks 2 (+0.36), with the other MLL3/MLL4-*specific* subunits behind (PAGR1 58, PAXIP1 84), while the core shared by all six SET1/MLL complexes is not enriched (RBBP5 1,052; WDR5 2,926).

## WHAT IT DOES NOT CLAIM

- **No causal direction.** Both measurements are correlational in the relevant sense; a transcriptional response cannot separate a direct target from a consequence of a shift in state.
- **Colorectal means HCT116 here.** Contrast agreement is +0.262 for one cell line across labs but +0.144 for different lines in one tissue, and cell line is confounded with laboratory — so three HCT116 datasets show the response reproduces *in HCT116*, not in colorectal cancer.
- **One line is flagged exploratory, with its p-value marked unusable.** KMT2D perturbation lowers ESRP1, ESRP2 and RBM47 and raises QKI and RBFOX2, while PTBP1 and MBNL1 barely move; the DepMap scan, run earlier and blind to those contrasts, separates the same six genes by seventeen thousand positions. But the five genes were chosen after seeing their fold changes, so the p-value is recorded as not usable.

## THREE CLAIMS MY OWN CONTROLS KILLED

- **That a KMT2D-dependent inflammatory programme exists** — the project's original premise. The derived signature contained no inflammatory genes. The line was closed and the reason recorded, not rewritten into a weaker version.
- **That each lineage has its own co-dependency profile.** Split-half self-agreement is +0.090 within lineage against +0.053 between — and colorectal self-agrees at +0.052, identical to the between-lineage figure. At 67–133 lines the ranking does not reproduce against itself. Claim withdrawn.
- **That screen depth undercut GRHL2.** I had compared two marginal correlations, which is not a test. The partial correlation puts GRHL2 at −0.127, *stronger* than its marginal value — while PARD3 really does collapse under the same control, from −0.073 to +0.002.

## METHOD

- **Pre-registration with an audit trail.** The plan and every revision, each with the result that forced it.
- **study_group, not sample, is the unit of evidence.** Contrasts sharing a cell line are not independent; cross-validation holds out whole groups.
- **Nulls matched to the question.** Arm relabelling within contrast; expression-matched random gene sets; split-half self-agreement before comparing groups.
- **Guards against failures that do not raise.** Identifier loss is measured, never silent; genes are intersected across series, not unioned.

`Python` `R` `salmon / tximport` `DepMap` `GEO` `pytest + CI`

Study design, statistical approach and interpretation are mine; implementation was AI-assisted. Code and full pre-registration available on request — the repository is private because it also references unpublished laboratory data.